

Evidence Based Clinical Practice Guidelines

Chapter 5

Introductions

- **Evidence-based clinical practice guidelines** are “systematically developed statements to assist practitioner and patient decisions about health care for specific circumstances.”
- Clinical practice guidelines are developed by a variety of groups and organizations including
 - Governmental institutions,
 - Professional societies and associations,
 - Managed care organizations,
 - Third party payers,
 - Quality assurance organizations &
 - Utilization review groups.

Introductions

- Many of the **skills required** for guideline development are required of **DI practitioners**, including:
 - Clear, specific definition of **clinical questions**,
 - **Literature** search & evaluation,
 - Epidemiology,
 - Clinical expertise
 - Writing, editing, formatting & education.
- Properly developed, valid practice guidelines provide **a concise summary of current best evidence** on specific health care interventions.

Introductions

- **Clinical practice guidelines:**
 - Assist the practitioner in **decision-making for specific pts** & also facilitate discussion of care options most consistent with individual pt needs & preferences.
 - **enhance provider communication** and **continuity of care**, especially when decisions are made by multiple providers in different care settings.
 - **help speed up the process of getting evidence into practice**
 - Assist with health care decision-making and to identify indicators for monitoring quality of care

Evidence-based medicine (EBM)

- EBM requires both clinical expertise & an intimate knowledge of the individual pt's situation, beliefs, priorities & values to be useful.
- ***External evidence must be used to inform, but not replace, individual clinical expertise.***
- It is clinical expertise that determines if the external evidence may be applied to the individual pt.

Evidence-based medicine (EBM)

- All health care professionals should be educated to deliver pt-centered care as members of an **interdisciplinary team, emphasizing evidence-based practice, quality improvement approaches & informatics.**
- However, they face the **complicated reality of constantly changing & increasing medical knowledge.**
- What is required to practice effective, high quality medicine is the **skills to acquire & critically assess** the specific information that is necessary to make clinical decisions.
- The decision-making process of EBM is supported by **access and use of clinical practice guidelines.**

- A thorough understanding of the methodology used for clinical practice guideline development is critical for pharmacists.
- Evaluation of the quality of a guideline & the appropriateness of its use in a given setting, depends primarily on an ability to distinguish methods that minimize **potential biases in dev't.**
- Although many organizations produce clinical practice guidelines, there are **no uniformly endorsed standards for their dev't.**

- The Institute of Medicine (**IOM**) proposed ***eight standards for developing evidence-based clinical practice guidelines***. The major steps in their proposed standards include the following:
 1. Establish transparency.
 2. Manage conflict of interest.
 3. Establish a multidisciplinary guideline development group.
 4. Conduct a systematic review for qualifying evidence.
 5. Establish evidence foundations for and rate strength of recommendations.
 6. Articulate recommendations.
 7. Conduct an external review.
 8. Establish a plan for updating the guideline.

1. Establish Transparency

- Guideline is developed and funded to be transparent, i.e., **how the guideline is developed and funded should be explicitly detailed and the public should have *access to this information*.**
- Transparency helps users understand how the recommendations were made and who developed them.
- The **clinical experience** of the guideline development group's members & **potential conflicts of interest** (COIs) should be stated as well.

2. Manage Conflict of Interest

- **Managing potential COIs by members of the guideline development group.** COIs not only include financial COIs, but also **intellectual conflicts** that may occur (previous research published).
- Individuals being considered for membership on a guideline development panel should be asked to declare potential COIs.
- Individuals with a potential COIs may still be considered for participation on a panel depending on the type & degree of conflict, along with appropriate **levels of management & disclosure.**
- Panel members who had COI could be **limited to some extent in their participation in discussions.**

3. Establish a Multidisciplinary Guideline Devn't Group

- The **composition** of the group to develop the guideline includes **patient** and **public involvement**.
- Patient involvement is especially important in **helping to formulate & prioritize the questions** to be addressed by the guideline.
- The development of a clinical practice guideline should involve a **multidisciplinary team**.
- Ideally, **all groups** that have a stake in the development & implementation of a guideline are represented in the process.

4. Conduct a Systematic Review for Qualifying Evidence

- The standard for this step acknowledges the **overlap between clinical practice guidelines and systematic reviews.**
- Systematic reviews are important for preparing a synthesis of evidence for the guideline development group to use in formulating recommendations
- This standard recommends that clinical practice guideline developers conduct systematic reviews that meet the Institute of Medicine (**IOM's**) **companion publication standard.**

4. Conduct a Systematic Review for Qualifying Evidence

- A few key steps in the **companion publication** for systematic reviews of comparative research include the following:
 - 1) Select an **appropriate topic for creation of a guideline.**
 - 2) Define **the clinical questions to be addressed.**
 - 3) Determine the **study screening selection criteria.**
 - 4) **Conduct a systematic search** for evidence.
 - 5) Critically **appraise** individual studies
 - 6) **Synthesize** the body of evidence

Select an Appropriate Topic for Creation of a Guideline

- Selection of a topic for guideline development has aspects in common with **selection of topics for a medication use evaluation program**, or in a broader sense for any **QI program**.
- As with a clinical management decision, the **potential benefits** of development and implementation of a guideline should be assessed.

Select an appropriate topic for creation of a guideline

- **Disease states** with the maximum potential for benefit from guideline development & implementation are:
 - 1) ***High prevalence***
 - 2) High ***frequency and/or severity of associated morbidity or mortality***
 - 3) Availability of ***high-quality evidence*** for the efficacy of treatments
 - 4) ***Feasibility*** of implementation of the treatment
 - 5) ***Potential cost-effectiveness***
 - 6) ***Evidence that current practice is not optimal***
 - 7) ***Evidence of practice variation***; similar pts are provided with different services or a substantially different frequency of services in different locations
 - 8) Availability of ***personnel, expertise & resources*** to develop and implement the practice guideline

Define the Clinical Questions to Be Addressed

- Define the specific issues for which recommendations will ultimately be provided.
- The ***decision-making points*** can be expressed as clinical questions.
- ***The definition of the clinical questions to be addressed by a guideline is a key step that provides direction for the activities to follow.***
- The questions are important to provide ***direction to the systematic review of the literature***, and they also provide the ***outline for the recommendations*** that the guideline will provide.

Define the Clinical Questions to Be Addressed

- Many guideline development groups use the **Patients-Interventions-Comparison-Outcomes (PICO) model**:
- **Patients**: Which pts are being considered for the guideline, how can they be described, and are there any subgroups that require special consideration?
- **Interventions**: Which intervention or treatment should be considered?
- **Comparison**: What other interventions or treatment should be compared with the intervention being considered?
- **Outcome**: What is most important to the patient (e.g., mortality, morbidity, treatment complications, rates of relapse, physical function, quality of life, and costs)?

- It is necessary to define the admissible evidence, i.e., ***the types of published or unpublished research to be considered*** so that an appropriate literature search may be performed.
- The guideline panel may decide to consider evidence from ***previous guidelines, meta-analyses or systematic reviews, RCT, observational studies and other studies***.
- ***Clear criteria*** must be used so that there is consistency among all individuals involved in this process.

Conduct a Systematic Search for Evidence

- Evidence-based guidelines require that all relevant evidence is located and appraised; therefore, a thorough literature search must be conducted.
- Many guideline development groups will *first conduct a search to identify previously completed guidelines or systematic reviews* of the same or closely related questions.
- The literature retrieval process should include a search of the available bibliographic resources such as **MEDLINE**[®], Current Contents[®], EMBASE, Science Citation Index, Cochrane Library, and Cumulative Index to Nursing and Allied Health Literature (CINAHL).

- There are a variety of methods for evaluating individual studies.
- The purpose of this process is ***to identify issues with the trial design or any biases that would affect internal or external validity.***
- Issues to consider include the ***basic trial design*** (i.e., randomized, controlled, clinical trial, cohort study, and case-control study), ***sample size, statistical power, selection bias, inclusion/exclusion criteria, choice of control group, randomization methods,*** comparability of groups, definition of exposure or intervention, definition of outcome measures, accuracy and appropriateness of outcome measures, attrition rates, data collection methods, methods of statistical analysis, confounding variables, unique characteristics of the study population, and adequacy of blinding.

Synthesize the Body of Evidence

- The evidence from the selected studies should be summarized in a **format** that facilitates the panel to begin developing conclusions.
- The best formats facilitate consideration of the characteristics and quality of individual studies, the consistency of the results between studies, the overall size of the evidence database, and the size of the treatment effects for benefits and harms.
- If the necessary evidence is available, it may be appropriate to *perform a meta-analysis* to present the summary estimate of the size of a treatment effect.

5. Establish *Evidence Foundations for* and *Rating Strength of Recommendations*

- Very important process of rating the *quality of the evidence and the strength of the recommendations* is produced.
- The Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) Working Group has developed the most accepted system for determining strength of recommendations and rating quality of evidence.
- The GRADE system, meets essential qualities of the IOM recommended standards.

5. Establish Evidence Foundations for and Rating Strength of Recommendations

GRADE System of Grading Quality of Evidence and Strength of Recommendations

- *The quality of the evidence that forms the **basis for recommendations** is a key aspect for interpretation and use of a practice guideline.*
- The system proposed by GRADE uses explicit definitions of what is meant by quality of evidence and strength of recommendation.
- **The quality of evidence** indicates the degree of confidence that an *estimate of effect is correct.*
- **The strength of a recommendation** indicates the degree of confidence that following the recommendation *will do more good than harm.*

5. Establish Evidence Foundations for and Rating Strength of Recommendations

- Use of the GRADE system requires assessment of the validity of the results of individual studies for important outcomes and sequential judgments about the following:
 - ❖ The ***quality of evidence across studies*** for each ***important outcome***
 - ❖ Which ***outcomes are critical*** to a decision
 - ❖ The overall quality of evidence across these ***critical outcomes***
 - ❖ The balance between ***benefits*** and ***harms***
 - ❖ The ***strength of the recommendations***

5. Establish Evidence Foundations for and Rating Strength of Recommendations

- The GRADE system starts with clearly defined clinical questions using the PICO model and considers all outcomes that are important to patients.
- The GRADE system classifies ***each outcome*** as
 - Critical,
 - Important but not critical, or
 - Of limited importance.
- Guideline developers may also use a 9-point scale with
 - 7-9 representing ***critical outcomes***,
 - 4-6 representing ***important outcomes***,
 - 1-3 representing ***outcomes of limited importance***.
- Critical outcomes*** are given more weight in the final recommendation.

5. Establish Evidence Foundations for and Rating Strength of Recommendations

- The ***quality of evidence*** for each outcome should be judged on the basis of a systematic review using **explicit criteria**.
- The GRADE system initially considers ***randomized trials*** to be **high-quality evidence** and ***observational studies*** to be ***low-quality evidence***.
- There are 5 reasons for ***lowering the level of confidence rating*** (risk of bias, inconsistency, indirectness, imprecision, publication bias), and
- There are 3 reasons for rating the level of ***confidence higher*** (large effect, dose response, all plausible confounding)

5. Establish Evidence Foundations for and Rating Strength of Recommendations

- Based on consideration of the components described above, the GRADE system arrives at one of the following ***grades of evidence*** that express the *degree of confidence* that the true effect is close to the estimate of the effect:
 - ❖ **High (A)** = very confident
 - ❖ **Moderate (B)** = moderately confident
 - ❖ **Low (C)** = limited confidence
 - ❖ **Very low (D)** = very little confidence

Quality of evidences

A (high quality evidence)	Consistent evidence from well performed randomized, controlled trials or overwhelming evidence of some other form. <i>Further research is unlikely to change our confidence</i> in the estimate of benefit and risk.
B (moderate quality evidence)	Evidence from randomized, controlled trials with important limitations (<i>inconsistent results, methodologic flaws, indirect or imprecise</i>), or very strong evidence of some other form. <i>Further research (if performed) is likely to have an impact on our confidence</i> in the estimate of benefit and risk and may change the estimate.
C and D (low quality evidence)	Evidence from observational studies, unsystematic clinical experience, or from randomized, controlled trials with serious flaws . Any estimate of effect is uncertain.

5. Establish Evidence Foundations for and Rating Strength of Recommendations

- Finally, the GRADE system assigns one of the following categories for a recommendation for an intervention:
 - ✍ Strong recommendation **for** using an intervention (1),
 - ✍ Weak recommendation **for** using an intervention (2),
 - ✍ Weak recommendation **against** an intervention (2), or
 - ✍ A strong recommendation **against** using an intervention (1)

5. Establish Evidence Foundations for and Rating Strength of Recommendations

- Factors involved in arriving at one of these four categories include the
 -  The balance of benefits to risk,
 -  The quality rating of the evidence,
 -  The variability in the values and preferences that may occur between different patients in different circumstances, and
 -  Resource use implications.

5. Establish Evidence Foundations for and Rating Strength of Recommendations

- In GRADE, grades have two components, a **number** (1 or 2) reflecting the ***strength of the recommendation*** and a **letter** (A, B, C, or D) reflecting the ***quality of the evidence*** supporting that recommendation.
- ***A Grade 1*** recommendation is a ***strong recommendation to do*** (or ***not do***) something, where the benefits clearly outweigh the risks (or vice versa) for most patients.
- ***A Grade 2*** recommendation is a ***weaker recommendation***, where the risks and benefits are more closely balanced or are more uncertain.

Grade of recommendation	Clarity of risk/ benefit	Implications
1A. Strong recommendation High quality evidence	Benefits clearly outweigh risk and burdens, or vice versa.	Strong recommendation, can apply to most patients in most circumstances without reservation.
1B. Strong recommendation Moderate quality evidence	Benefits clearly outweigh risk and burdens, or vice versa.	Strong recommendation, likely to apply to most patients.
1C. Strong recommendation Low quality evidence	Benefits appear to outweigh risk and burdens, or vice versa.	Relatively strong recommendation; might change when higher quality evidence becomes available.

2A. Weak recommendation High quality evidence	Benefits closely balanced with risks and burdens.	Weak recommendation, best action may differ depending on circumstances or patients or societal values.
2B. Weak recommendation Moderate quality evidence	Benefits closely balanced with risks and burdens, some uncertainty in the estimates of benefits, risks and burdens.	Weak recommendation, alternative approaches likely to be better for some pts under some circumstances.
2C. Weak recommendation	Uncertainty in the estimates of benefits, risks, and burdens; benefits may be closely	Very weak recommendation; other alternatives may be equally reasonable.

6. Articulate Recommendations

- This step addresses the details of how the guideline recommendations should be *phrased, specifying what actions should be taken* and in what **settings and circumstances**.
- *Standardized language* should be used to detail exactly what the recommendation is and when it should be done.
- When justified by the evidence, these recommendations should be ***clear and concise with statements that are actionable*** and ***employ active voice***, with avoidance of vague and nonspecific language

6. Articulate Recommendations

- Following the GRADE system a ***strong recommendation*** would be introduced by terms such as “**We recommend**” or “**Clinicians should**” and a ***weak recommendation*** would begin with the terms “We suggest,” “**Clinicians might,**” or “**Clinicians may.**”
- The basis of the draft document is provided by the graded recommendations.
- ***A formal narrative summary*** should also be provided with all relevant details of decisions made in the development of the guideline

6. Articulate Recommendations

- It is highly desirable for the finished guideline to include details of the scope of the guideline including:
 -  target patient population,
 -  restrictions on the population,
 -  interventions considered,
 -  specific outcomes or performance measures,
 -  who are the intended users of the guideline (e.g., specialty and care setting), and
 -  the overall objective of the guideline.
- **Quick reference guides**, which give the essentials of the recommendations without the detailed background are usually produced.

7. Conduct an External Review

- The standard for this step details how the guideline should undergo external review, categories for types of stakeholders and participants to be involved in the review, and calls for a draft to be made available for general public comment.
- All relevant stakeholders should be involved in the external review.
- Based on feedback from external review, revisions may be required for the guideline to meet its intended goals.

8. Establish a plan for updating the guideline

- The standard for this step addresses the need to keep guidelines up to date with new information as it becomes available.
- It calls for clarity of the publication date, dates of systematic reviews utilized in the guideline, regular review of the literature to identify new technology or new evidence that may impact the guideline recommendation, and a specific plan for updating or expiration of a guideline
- A review interval to update the guideline should be established by the panel

Guideline Evaluation Tools

- Prior to selecting a clinical practice guideline for implementation in a health care system or for personal use by a health care professional, it is important that the quality of published guidelines be evaluated.
- Perhaps the most useful tool available for evaluating clinical practice guidelines is the one created by the Appraisal of Guidelines for Research and Evaluation (**AGREE**) collaboration.

Guideline Evaluation Tools

- The AGREE collaboration is an international group of researchers and policy makers.
- This collaboration has produced a structured instrument which can be used for critical appraisal of clinical practice guidelines.
- The collaboration refined the original instrument, which has resulted in the **AGREE II instrument** (structured, reliable, and reasonable tool).
- The purpose of the **AGREE II instrument** is to
 - provide a **framework to assess the quality of guidelines**,
 - **provide a methodological strategy for the development of guidelines**, &
 - Inform what and how information ought to be reported in guidelines

Implementation of Clinical Practice Guidelines

- *The most effective methods for implementing guidelines to achieve the desired effects of improved quality of care have not been determined.*
- *Institutional, organizational, local practice, political characteristics, and even individual practitioner characteristics should be considered when planning an implementation strategy for a practice guideline.*

Implementation of Clinical Practice Guidelines

- **Barriers** to guideline implementation should be considered when making plans.
- A barrier was defined as any factor that limits or restricts complete physician adherence to a guideline.
- Paying appropriate attention to these potential barriers in the planning and development of guidelines will facilitate successful implementation.

There are seven general categories of identified barriers.

Lack of awareness	Did not know the guideline existed
Lack of familiarity	Could not correctly answer questions about guideline content or self-reported lack of familiarity
Lack of agreement	Difference in interpretation of the evidence
Lack of self-efficacy	Did not believe that they could actually perform the behavior or activity recommended by the guideline, e.g., nutrition or exercise counseling
Lack of outcome expectancy	Did not believe intended outcome would occur even if the practice was followed, e.g., counseling to stop smoking
Inertia of previous practice	This barrier relates primarily to motivation to change practice, whether the motivation is professional, personal, or social.
External barriers	Patient resistance/nonadherence, cost, lack of time

Sources of Clinical Practice Guidelines

- Complete clinical practice guidelines can be found on **Web sites** such as *the National Guideline Clearinghouse (NGC)* (<http://www.guideline.gov>)
 - The NGC has published criteria for guidelines to be considered for inclusion in the Clearinghouse (<http://www.guideline.gov/about/inclusion-criteria.aspx>).
 - Many guidelines have been published *the peer-reviewed medical literature located by use of secondary databases such as MEDLINE®*

Sources of Clinical Practice Guidelines

- The National Institutes of Health (NIH) consensus statements, the U.S. Preventative Services Task Force (USPSTF) guides, AHRQ evidence reports & summaries, Web site which is published by the *National Library of Medicine* (<http://www.ncbi.nlm.nih.gov/books/NBK16710>)
- The Guidelines International Network (G-I-N) (<http://www.g-i-n.net>) provides access to the International Guideline Library.

Sources of Clinical Practice Guidelines

- Clinical practice guideline can also be obtained by direct contacting or via webpages of
 - multiple professional organizations, academic centers, independent research centers, and government agencies
- Systematic reviews available at Cochrane Collaboration databases (<http://www.cochrane.org/about-us/evidence-based-health-care/webliography/databases>).

Sources of Clinical Practice Guidelines

- American Diabetic Association <https://www.diabetes.org/>
- American College of Cardiology <https://www.acc.org/>
- AIDSInfo Clinical Guidelines
<http://aidsinfo.nih.gov/guidelines>
- American Academy of Neurology
<http://www.aan.com/go/practice/guidelines>
- SIGN: Scottish Intercollegiate Guidelines Network
<http://www.sign.ac.uk>
- Evidence-based Practice
<http://www.health.uottawa.ca/rehabguidelines/>

Local sources

Local websites

- Federal Ministry of Health: www.moh.gov.et
- Ethiopian Food and drug administration www.EFDA.gov.et